



9/1/2026

N + - Network Fundamentals & Building Networks.

OSI Layer.

TASK 1.

❖ Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

- **OSI 7-Layer Model:** The OSI model divides network communication into seven layers, with each layer performing a specific function.

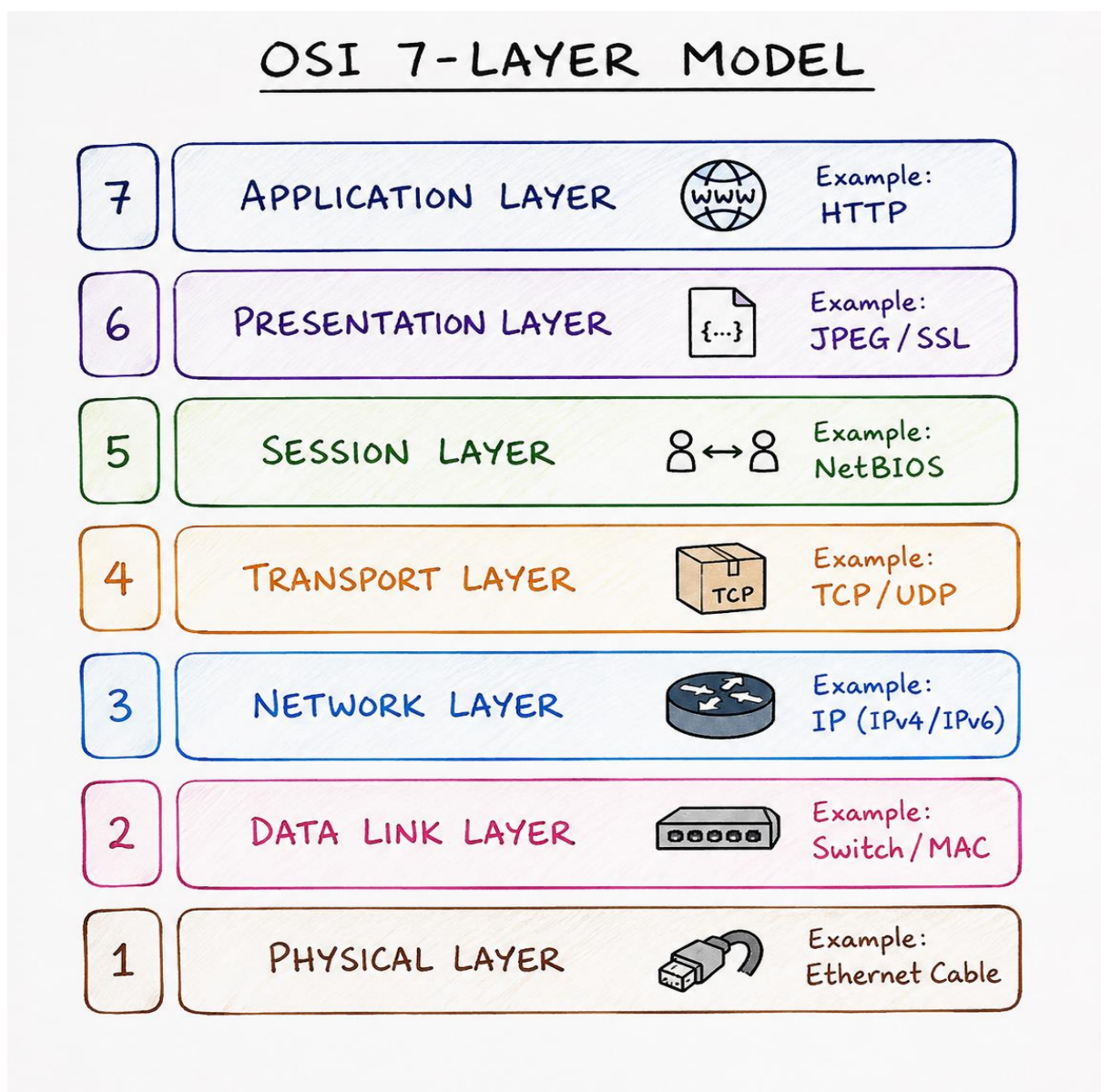


Figure : OSI 7-Layer Model.

⌘ Layer 7 – Application Layer

⇒ **Function:** Provides network services to user applications.

⇒ **Example:** HTTP – used for web communication.

⌘ Layer 6 – Presentation Layer

⇒ **Function:** Translates, encrypts, and compresses data.

⇒ **Example:** JPEG / SSL – used for data formatting and encryption.

⌘ Layer 5 – Session Layer

⇒ **Function:** Establishes, manages, and terminates communication sessions.

⇒ **Example:** NetBIOS – manages network sessions.

⌘ Layer 4 – Transport Layer

⇒ **Function:** Provides end-to-end data delivery and controls data flow.

⇒ **Example:** TCP / UDP – transport protocols.

⌘ Layer 3 – Network Layer

⇒ **Function:** Handles logical addressing and routing of data.

⇒ **Example:** IP (IPv4/IPv6) – used for addressing and routing.

⌘ Layer 2 – Data Link Layer

⇒ **Function:** Transfers data between connected devices and detects errors.

⇒ **Example:** Switch / MAC Address – used for device-to-device communication.

⌘ **Layer 1 – Physical Layer**

- ⇒ **Function:** Transmits raw bits through physical media.
- ⇒ **Example:** Ethernet Cable – carries signals between devices.

TASK 2.

❖ **Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames, Write your answers as 'Scenario: Layer Name (Layer Number)'.**

➤ **The correct OSI layer for each networking scenario is given below:**

- ⇒ Encrypting data before sending: **Presentation Layer (Layer 6)**
- ⇒ Routing packets across networks: **Network Layer (Layer 3)**
- ⇒ Converting data to electrical signals: **Physical Layer (Layer 1)**
- ⇒ Opening a web page in your browser: **Application Layer (Layer 7)**
- ⇒ Detecting and correcting errors in frames: **Data Link Layer (Layer 2)**

TASK 3.

❖ Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

➤ OSI Layers Involved in WhatsApp:

↪ I use WhatsApp daily to send messages, photos, and other data. Different OSI layers work together to make this communication possible. Three important layers involved are:

⌘ **Application Layer (Layer 7):**

This layer provides services directly to the WhatsApp application. It handles activities such as sending and receiving messages, photos, and files.

⌘ **Transport Layer (Layer 4):**

This layer manages the reliable transfer of data between the sender and receiver. It helps ensure that the data reaches the correct destination.

⌘ **Network Layer (Layer 3):**

This layer handles IP addressing and routing. It decides how the data packets should travel across networks to reach the receiver.

🌈 **Conclusion:**

These OSI layers work together to ensure that data sent through WhatsApp reaches the intended user correctly and efficiently.

TASK 4.

❖ **Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**

➤ **The problem is most likely at the Application Layer (Layer 7) or Transport Layer (Layer 4).**

⇒ Since the IP address can be pinged successfully, the **Network Layer (Layer 3)** is working and basic connectivity is available. The issue may be with the web service, such as **HTTP/HTTPS**, or with the transport connection required to access the website.

⇒ Therefore, the problem is likely in a higher OSI layer, especially **Layer 4 or Layer 7**, rather than the basic network connection.